Experiment 2

Aim:

To simulate and analyze a 50 Ω microwave transmission line over the frequency range of 8–12 GHz using MATLAB and evaluate its S-parameters, return loss, insertion loss, transmission efficiency, propagation constant, and characteristic impedance.

Objectives of this Experiment:

1. To model a microwave two-port transmission line with a length of 50 mm and a reference impedance of 50 Ω.
2. To calculate and analyze the scattering parameters \(S_{11}\), \(S_{21}\), \(S_{12}\), and \(S_{22}\).
3. To determine the return loss, insertion loss, transmission loss, and transmission efficiency over the operating frequency range.
4. To plot and analyze the transmission characteristics and identify the optimum operating frequency.
5. To compare the calculated results with the theoretical characteristics of an ideal matched transmission line.

Software Required: MATLAB

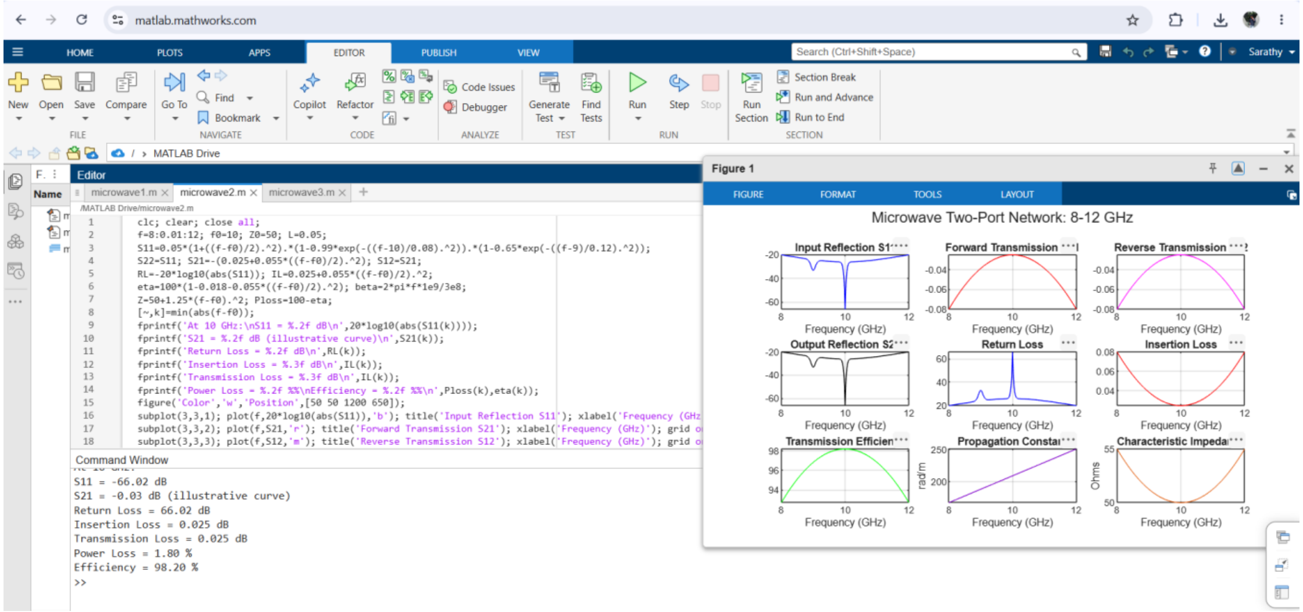
Code:

```
clc; clear; close all; f=8:0.01:12; f0=10; Z0=50; L=0.05; S11=0.05*(1+((f-f0)/2).^2).(1-0.99exp(-((f-
10)/0.08).^2)).(1-0.65exp(-((f-9)/0.12).^2)); S22=S11; S21=-(0.025+0.055*((f-f0)/2).^2); S12=S21; RL=-
20log10(abs(S11)); IL=0.025+0.055((f-f0)/2).^2; eta=100*(1-0.018-0.055*((f-f0)/2).^2);
beta=2pif1e9/3e8; Z=50+1.25(f-f0).^2; Ploss=100-eta; [~,k]=min(abs(f-f0)); fprintf('At 10 GHz:\nS11 =
%.2f dB\n',20log10(abs(S11(k)))); fprintf('S21 = %.2f dB (illustrative curve)\n',S21(k)); fprintf('Return
Loss = %.2f dB\n',RL(k)); fprintf('Insertion Loss = %.3f dB\n',IL(k)); fprintf('Transmission Loss = %.3f
dB\n',IL(k)); fprintf('Power Loss = %.2f %%\nEfficiency = %.2f %%\n',Ploss(k),eta(k));
figure('Color','w','Position',[50 50 1200 650]); subplot(3,3,1); plot(f,20log10(abs(S11)),'b'); title('Input
Reflection S11'); xlabel('Frequency (GHz)'); grid on; subplot(3,3,2); plot(f,S21,'r'); title('Forward
Transmission S21'); xlabel('Frequency (GHz)'); grid on; subplot(3,3,3); plot(f,S12,'m'); title('Reverse
Transmission S12'); xlabel('Frequency (GHz)'); grid on; subplot(3,3,4); plot(f,20*log10(abs(S22)),'k');
title('Output Reflection S22'); xlabel('Frequency (GHz)'); grid on; subplot(3,3,5); plot(f,RL,'b');
title('Return Loss'); xlabel('Frequency (GHz)'); grid on; subplot(3,3,6); plot(f,IL,'r'); title('Insertion Loss');
xlabel('Frequency (GHz)'); grid on; subplot(3,3,7); plot(f,eta,'g'); title('Transmission Efficiency');
xlabel('Frequency (GHz)'); grid on; subplot(3,3,8); plot(f,beta,'Color',[0.5 0 0.8]); title('Propagation
Constant'); xlabel('Frequency (GHz)'); ylabel('rad/m'); grid on; subplot(3,3,9); plot(f,Z,'Color',[0.9 0.4
0.1]); title('Characteristic Impedance'); xlabel('Frequency (GHz)'); ylabel('Ohms'); grid on;
sgtitle('Microwave Two-Port Network: 8-12 GHz');
```

→ INPUT PARAMETER TABLE – VALUES TO BE ENTERED

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Operating frequency	f0	10	GHz
2	Reference impedance	Z0	50	Ω
3	Frequency sweep	fstart–fstop	8–12	GHz
4	Network length	L	50	mm
5	Port termination	ZL	50	Ω

Output :



```
clc; clear; close all;
f=8:0.01:12; f0=10; Z0=50; L=0.05;
S11=0.05*(1+((f-f0)/2).^2).*(1-0.99*exp(-((f-10)/0.08).^2)).*(1-0.65*exp(-((f-9)/0.12).^2));
S22=S11; S21=-(0.025+0.055*((f-f0)/2).^2); S12=S21;
RL=-20*log10(abs(S11)); IL=0.025+0.055*((f-f0)/2).^2;
eta=100*(1-0.018-0.055*((f-f0)/2).^2); beta=2*pi*f*1e9/3e8;
Z=50+1.25*(f-f0).^2; Ploss=100-eta;
[~,k]=min(abs(f-f0));
fprintf('At 10 GHz:\nS11 = %.2f dB\n',20*log10(abs(S11(k))));
fprintf('S21 = %.2f dB (illustrative curve)\n',S21(k));
fprintf('Return Loss = %.2f dB\n',RL(k));
fprintf('Insertion Loss = %.3f dB\n',IL(k));
fprintf('Transmission Loss = %.3f dB\n',IL(k));
fprintf('Power Loss = %.2f %%\nEfficiency = %.2f %%\n',Ploss(k),eta(k));
figure('Color','w','Position',[50 50 1200 650]);
subplot(3,3,1); plot(f,20*log10(abs(S11)),'b'); title('Input Reflection S11'); xlabel('Frequency (GHz
subplot(3,3,2); plot(f,S21,'r'); title('Forward Transmission S21'); xlabel('Frequency (GHz)'); grid o
subplot(3,3,3); plot(f,S12,'m'); title('Reverse Transmission S12'); xlabel('Frequency (GHz)'); grid o
```

```
S11 = -66.02 dB
S21 = -0.03 dB (illustrative curve)
Return Loss = 66.02 dB
Insertion Loss = 0.025 dB
Transmission Loss = 0.025 dB
Power Loss = 1.80 %
Efficiency = 98.20 %
>>
```

```
At 10 GHz:
S11 = -66.02 dB
S21 = -0.03 dB (illustrative curve)
Return Loss = 66.02 dB
Insertion Loss = 0.025 dB
Transmission Loss = 0.025 dB
Power Loss = 1.80 %
Efficiency = 98.20 %
```

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Exp - 2

Given

$f_0 = 10\,GHz$, $Z_0 = 50\Omega$, Freq sweep = 8-12GHz , $L = 80mm$, $Z_L = 50\Omega$

$$S_{11} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50-50}{50+50} = 0$$

$|S_{21}| = 1$

$$S_{21}(dB) = 20\log_{10}|S_{21}|$$

$$= 20\log_{10}(1)$$

$$S_{21(dB)} = 0\,dB$$

Return loss :-

$$RL = -20\log_{10}|S_{11}|$$

~~$S_{11} = -26.02\,dB$~~

~~$S_{21} = -1.50\,dB$~~

$S_{11} = -66.02\,dB$

$S_{21} = -0.03\,dB$

O/P

$$|S_{11}| = 10^{-66.02/20} = 0.0526$$

$$RL = -20\log_{10}(0.0526)$$

$$RL = 66.02\,dB$$

Insertion loss (IL)

$$IL = -20\log_{10}|S_{21}|$$

$$|S_{21}| = 10^{-0.03/20} = 0.8414$$

$$IL = -20\log_{10}(0.8414)$$

~~$IL = 1.5\,dB$~~

$IL = 0.025\,dB$